

CS 660: Data Mining For Decision Making

Lecture 24 (Week 11)

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Scaling the Heights

Compare Models in Their Ability To Explain Data

- Comparisons based upon:
 - MSD/Chi-squared
 - MSD = Sum over data [Model proportion – Human proportion]² / Total problems)
 - Chi-squared: Sum over Bins [Model Frequency – Human Frequency]² / [Human Frequency]; Frequency is counts of the dependent (predicted) variable in the bin.
 - AIC/BIC
 - AIC = $G^2 + 2 * k$; k: number of free parameters
 - BIC = $\frac{1}{2} * (G^2 + 2 * k * \ln(N))$; N: number of trials
 - $G^2 = N * \ln(SSE/N)$ (for aggregate, hierarchical)
 - $G^2 = -2 * \ln(L)$ (for individual)
 - L: Likelihood Estimator (used for individual):
 - SSE = $\sum_{i=1}^t (x_{\text{model},i} - x_{\text{human},i})^2$
 - probability of picking an option from the model across all trials by a participant in a problem

Likelihood Estimator (L)

- $P(\text{Data} \mid \text{Model and its parameters})$: Probability of predicting the data given the model and its parameters

Participant	Problem	Round	Play (1) or Not Play (0)	Prediction From a Model
1	1	1	1	0.5
1	1	2	0	0.5
1	1	3	1	0.6
...	...	...	...	...
1	1	100	0	0.3

- The $L = 0.5 \cdot 0.5 \cdot 0.6 \dots \cdot 0.3$; we define $\ln L = \ln(0.5) + \ln(0.5) + \dots + \ln(0.3)$; $G^2 = -2 \cdot \ln L$ and the idea is to calibrate parameters to maximize $\ln L$ or minimize the G^2

Chi-Squared

- Excel Example!

Thank you!

Comments and Questions most welcome!